

Linear Algebra Project

How to use LU Factorization in real-life?

GROUP MEMBERS

Beyza Miyanyedi

Esmâ Uğurlı

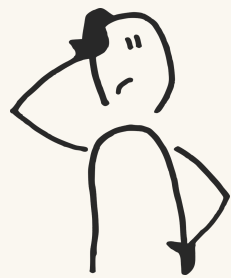
Hale Nesibe Köse

Semanur Topal

Şeyma Babacan

INTRODUCTION

Project Overview



Problem

Machine capacities are fixed, while customer demands constantly change. Since products share common resources, every new demand affects the entire system — and in large-scale operations, solving everything from scratch each time is both slow and prone to numerical errors.



Solution

The matrix A , which represents the factory's structure, is decomposed only once using LU Factorization. When demand changes, the system is not re-solved from scratch; instead, forward and backward substitution are applied to the new demand vector only. This reduces computational complexity from $O(n^3)$ to $O(n^2)$.



Why It Matters

Linear algebra stops being abstract and becomes the real-time decision engine of a factory. Managers can simulate different demand scenarios instantly, detect bottlenecks early, and minimize operational costs.

INTRODUCTION

Project Objective

Main Goal

The main goal of this project is to demonstrate how linear algebra concepts — specifically LU Factorization — can be applied to a real-world industrial engineering problem. By modeling factory production as a linear equation system, we aim to build a fast, efficient, and interactive system that helps production managers adapt to changing market demands in real time.



INTRODUCTION

Problem Definiton

Problem

A factory produces three products on three machines. Each product requires different processing times on each machine, and each machine has a fixed capacity. As customer demands change continuously, the goal is to find the optimal production quantities that fully meet demand without exceeding machine capacities — by modeling the system as a linear equation system $Ax = b$ and solving it efficiently using LU Factorization.



Mathematical Implementation and Theoretical Background

The factory produces three products using three machines. Producing one unit of Product 1 requires 2 hours on Machine 1, 4 hours on Machine 2, and 2 hours on Machine 3. One unit of Product 2 requires 1 hour on Machine 1, 3 hours on Machine 2, and 2 hours on Machine 3. Finally, one unit of Product 3 requires 1 hour on Machine 1, 2 hours on Machine 2, and 2 hours on Machine 3. The available capacities are 100 hours for Machine 1, 120 hours for Machine 2, and 110 hours for Machine 3. These constraints form the linear equation system $Ax = b$, where x_1 , x_2 , x_3 represent the production quantities and d_1 , d_2 , d_3 represent the capacity limits of each machine. (THIS PART IS FROM OUR REPORT)

To save time, we will not include how to factorize a matrix, but to briefly summarize our operations, we are sharing the factorized matrices and the step-by-step solution with you. More detailed mathematical procedures are provided in our report.

Mathematical Implementation and Theoretical Background

Matrix A:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

Matrix L(Lower
Triangular Matrix):

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ 0.5 & -1 & 1 \end{bmatrix}$$

Matrix U(Upper
Triangular Matrix):

$$U = \begin{bmatrix} 4 & 3 & 2 \\ 0 & -0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix
P(Permutation
Matrix):

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Mathematical Implementation and Theoretical Background

1

Calculate Pd , where d is the demand vector.

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, d = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$
$$Pd = \begin{bmatrix} d_2 \\ d_1 \\ d_3 \end{bmatrix}$$

2

Calculate $Ly = Pd$

$$y_1 = d_2$$
$$y_2 = d_1 - 0.5d_2$$
$$y_3 = d_1 - d_2 + d_3$$

3

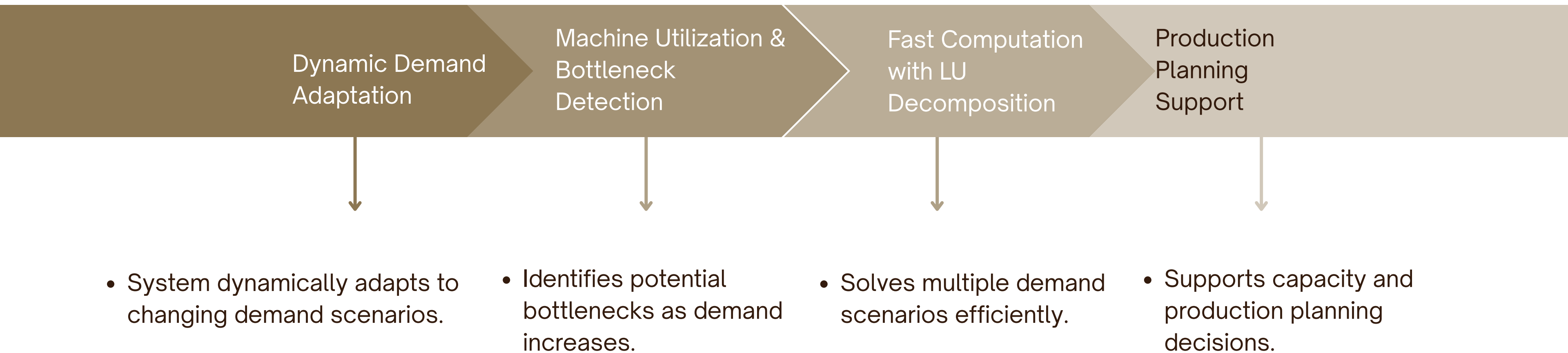
Calculate $Ux=y$

$$x_3 = y_3$$
$$x_2 = -2y_2$$
$$x_1 = \frac{y_1 - 3x_2 - 2x_3}{4}$$

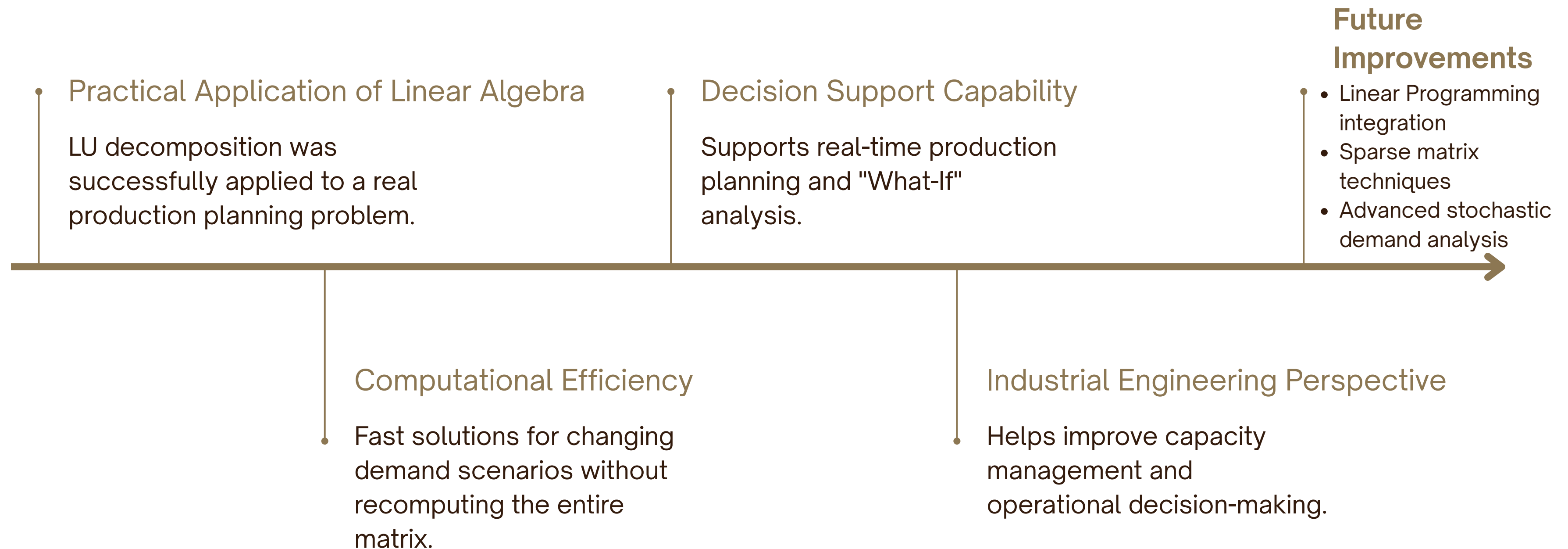
Our Dashboard Structure



RESULTS & ANALYSIS



CONCLUSION



The End

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